

MTH313M End Semester Exam Part A Answers

Question 1. (Set 1) Consider an $M/M/1$ queueing process with arrival rate $\lambda = 1$ per unit of time and service distribution with rate $\mu = 2$ per unit of time. Assume that the system has infinite capacity.

- (a) (2 marks) In the steady-state of the system, the average number of customers in the system is $\frac{\lambda}{\mu - \lambda} = 1$
- (b) (2 marks) In the steady-state of the system, the average time spent by a customer in the system is $\frac{1}{\mu - \lambda} = 1$ unit
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is $\frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{1}{2}$
- (d) (3 marks) In the steady-state of the system, the probability that an arriving customer will find the the server to be idle is $1 - \frac{\lambda}{\mu} = \frac{1}{2}$

(Set 2) Consider an $M/M/1$ queueing process with arrival rate $\lambda = 1$ per unit of time and service distribution with rate $\mu = 3$ per unit of time. Assume that the system has infinite capacity.

- (a) (2 marks) In the steady-state of the system, the average number of customers in the system is $\frac{\lambda}{\mu - \lambda} = \frac{1}{2}$
- (b) (2 marks) In the steady-state of the system, the average time spent by a customer in the system is $\frac{1}{\mu - \lambda} = \frac{1}{2}$ unit
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is $\frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{1}{6}$
- (d) (3 marks) In the steady-state of the system, the probability that an arriving customer will find the the server to be idle is $1 - \frac{\lambda}{\mu} = \frac{2}{3}$

(Set 3) Consider an $M/M/1$ queueing process with arrival rate $\lambda = 2$ per unit of time and service distribution with rate $\mu = 3$ per unit of time. Assume that the system has infinite capacity.

- (a) (2 marks) In the steady-state of the system, the average number of customers in the system is $\frac{\lambda}{\mu - \lambda} = 2$
- (b) (2 marks) In the steady-state of the system, the average time spent by a customer in the system is $\frac{1}{\mu - \lambda} = 1$ unit
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is $\frac{\lambda^2}{\mu(\mu - \lambda)} = \frac{4}{3}$
- (d) (3 marks) In the steady-state of the system, the probability that an arriving customer will find the the server to be idle is $1 - \frac{\lambda}{\mu} = \frac{1}{3}$

Question 2. (Set 1) Consider an $M/M/1$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 2$ per unit of time. Assume that the system can accommodate upto 4 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (3 marks) In the steady-state of the system, the probability distribution $(p_0, p_1, p_2, p_3, p_4)$ of the number of customers in the system equals $(\frac{1}{31}, \frac{2}{31}, \frac{4}{31}, \frac{8}{31}, \frac{16}{31})$

Hint:

$$p_0 = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^5}$$

$$p_1 = \left(\frac{\lambda}{\mu}\right) \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^5}$$

$$p_2 = \left(\frac{\lambda}{\mu}\right)^2 \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^5}$$

$$p_3 = \left(\frac{\lambda}{\mu}\right)^3 \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^5}$$

$$p_4 = \left(\frac{\lambda}{\mu}\right)^4 \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^5}$$

- (b) (2 marks) In the steady-state of the system, the average number of customers in the system is $\frac{98}{31}$

Hint:

$$L = \frac{1 - \frac{\lambda}{\mu}}{1 - \left(\frac{\lambda}{\mu}\right)^5} \sum_{j=1}^4 j \left(\frac{\lambda}{\mu}\right)^j$$

- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is $\frac{68}{31}$

$$\text{Hint: } \sum_{j=1}^4 (j-1)p_j = \sum_{j=1}^4 jp_j - \sum_{j=1}^4 p_j = L - (1 - p_0) = L + p_0 - 1$$

(Set 2) Consider an $M/M/1$ queueing process with arrival rate $\lambda = 3$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time. Assume that the system can accommodate upto 4 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (3 marks) In the steady-state of the system, the probability distribution $(p_0, p_1, p_2, p_3, p_4)$ of the number of customers in the system equals $\left(\frac{1}{121}, \frac{3}{121}, \frac{9}{121}, \frac{27}{121}, \frac{81}{121}\right)$
- (b) (2 marks) In the steady-state of the system, the average number of customers in the system is $\frac{426}{121}$

- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is $\frac{306}{121}$

(Set 3) Consider an $M/M/1$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time. Assume that the system can accommodate upto 4 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (3 marks) In the steady-state of the system, the probability distribution $(p_0, p_1, p_2, p_3, p_4)$ of the number of customers in the system equals $\left(\frac{1}{341}, \frac{4}{341}, \frac{16}{341}, \frac{64}{341}, \frac{256}{341}\right)$
- (b) (2 marks) In the steady-state of the system, the average number of customers in the system is $\frac{1252}{341}$
- (c) (2 marks) In the steady-state of the system, the average number of customers waiting in the queue is $\frac{912}{341}$

Question 3. Consider an $M/M/4$ queueing process with arrival rate $\lambda = 4$ per unit of time and exponential service distribution with rate $\mu = 1$ per unit of time per server. Assume that the system can accommodate upto 6 customers at a time. It is understood that customers arriving to find the queue full leave immediately and are not considered to be in the system.

- (a) (1 mark) Think of this queueing process as a Birth and Death process. If there are two customers in the system at some time, the effective death rate of the system at this time is 2
- (b) (1 mark) Think of this queueing process as a Birth and Death process. If one customer is waiting in the queue at some time, the effective death rate of the system at this time is 4
- (c) (1 mark) In this queueing process, if one server becomes inoperable, can the system reach a steady state? Underline the correct answer. Yes/No

Question 4. Let $\{B_t\}_{t \geq 0}$ denote a standard Brownian motion.

- (a) (3 marks) Write down the distribution of
 - Set 1: $\begin{pmatrix} 1 + B_3 \\ B_5 - B_2 \end{pmatrix} \sim N \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix} \right)$
 - Set 2: $\begin{pmatrix} 1 + B_2 \\ B_4 - B_1 \end{pmatrix} \sim N \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix} \right)$
 - Set 3: $\begin{pmatrix} 2 + B_3 \\ B_4 - B_1 \end{pmatrix} \sim N \left(\begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \right)$
- (b) (2 marks) The sequence of random variables $\left\{ \frac{B_n}{n} : n = 1, 2, \dots \right\}$ converges almost surely to a random variable X . Then X is degenerate at 0
- (c) (2 marks)
 - Set 1: $\mathbb{E}|B_2|^3 = \frac{8}{\sqrt{\pi}}$
 - Set 2: $\mathbb{E}|B_3|^3 = \frac{6\sqrt{6}}{\sqrt{\pi}}$
 - Set 3: $\mathbb{E}|B_5|^3 = \frac{10\sqrt{10}}{\sqrt{\pi}}$
- (d) (1 mark) With probability 1, are the paths of $\{B_t\}_{t \in [0,1]}$ locally Hölder continuous of order $\frac{1}{3}$? Underline the correct answer. Yes/No
- (e) (2 marks) Consider the process $\{X_t\}_{t \geq 0}$ defined by $X_t = \frac{1}{2}(B_{4t+1} - B_1)$. Is $\{X_t\}_{t \geq 0}$ a standard Brownian motion? Underline the correct answer. Yes/No

Question 5. Let $X_t = \sum_{j=1}^{N_t} Y_j, t \geq 0$ be a Compound Poisson Process with rate $\lambda > 0$, where $Y_1, Y_2, \dots$ are i.i.d. $N(1, 3)$ random variables, independent of $\{N_t\}_{t \geq 0}$.

- (a) (1 mark) Does the process $\{X_t\}_{t \geq 0}$ have independent increments? Underline the correct answer. Yes/No
- (b) (1 mark) Does the process $\{X_t\}_{t \geq 0}$ have stationary increments? Underline the correct answer. Yes/No
- (c) (1 mark)
 - (Set 1) $\mathbb{E}X_t = \lambda t$ (for $N(1, 3)$)
 - (Set 2) $\mathbb{E}X_t = \lambda t$ (for $N(1, 5)$)
 - (Set 3) $\mathbb{E}X_t = 2\lambda t$ (for $N(2, 1)$)
- (d) (1 mark) The variance of X_t is
 - (Set 1) $4\lambda t$ (for $N(1, 3)$)
 - (Set 2) $6\lambda t$ (for $N(1, 5)$)

(Set 3) $5\lambda t$ (for $N(2, 1)$)

(e) (1 mark) For $u \in \mathbb{R}$,

(Set 1) $\mathbb{E}e^{iuX_t} = \exp(-\lambda t(1 - \exp(iu - \frac{3}{2}u^2)))$ (for $N(1, 3)$)

(Set 2) $\mathbb{E}e^{iuX_t} = \exp(-\lambda t(1 - \exp(iu - \frac{5}{2}u^2)))$ (for $N(1, 5)$)

(Set 3) $\mathbb{E}e^{iuX_t} = \exp(-\lambda t(1 - \exp(2iu - \frac{1}{2}u^2)))$ (for $N(2, 1)$)

Question 6. Let $\{N_t\}_{t \geq 0}$ denote a Poisson process with rate $\lambda > 0$.

(a) (2 marks) Write down the Kolmogorov's Forward Equation for this process.

$$p'_n(t) = \lambda p_{n-1}(t) - \lambda p_n(t), \quad n = 1, 2, \dots$$

and

$$p'_0(t) = -\lambda p_0(t)$$

and

$$p_0(0) = 1, p_n(0) = 0, \quad n = 1, 2, \dots$$

Note: 1) without the initial condition, the KFE for PP is incomplete

2) equivalent description is available with $p_{i,j}$

(b) (2 marks) Write down the Kolmogorov's Backward Equation for this process.

$$p'_n(t) = \lambda p_{n-1}(t) - \lambda p_n(t), \quad n = 1, 2, \dots$$

and

$$p'_0(t) = -\lambda p_0(t)$$

and

$$p_0(0) = 1, p_n(0) = 0, \quad n = 1, 2, \dots$$

Note: 1) without the initial condition, the KBE for PP is incomplete

2) equivalent description is available with $p_{i,j}$

(c) (1 mark) Are the Kolmogorov's Forward and Backward equations identical for a Poisson process? Underline the correct answer. Yes/No (Instruction: The sub-question (c) will only be graded if above two parts (a) and (b) have been answered.)

Question 7. Consider a Birth and Death process $\{X_t\}_{t \geq 0}$ with infinitesimal birth parameters $\lambda_0 > 0, \lambda_1 > 0, \lambda_2 > 0, \dots$ and infinitesimal death parameters $\mu_0 = 0, \mu_1 > 0, \mu_2 > 0, \dots$.

(a) (3 marks) Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$, the conditional probability that a birth occurs before a death is $\frac{\lambda_i}{\lambda_i + \mu_i}$

(b) (2 marks) Given that $X_t = i$ for some time $t \geq 0$ and integer $i > 0$. Let T denote the waiting time for either a birth or a death to occur. Then, $\mathbb{E}T = \frac{1}{\lambda_i + \mu_i}$